Strategic use of English quantifiers in the reporting of quantitative information

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ABSTRACT

This study investigates how quantifiers can be used strategically to serve different argumentative goals. We report two experiments on how English speakers describe the results of school exams when being instructed to frame their descriptions either as a good or bad outcome. Experiment 1 shows that participants have clear preferences for specific quantifier combinations. Experiment 2 shows that, in situations where producing truthful descriptions is difficult (i.e., framing very bad outcomes positively), participants tend to use quantifiers which are informationally weaker than other salient alternatives in context. Experiment 2 also shows that people have a bias to frame outcomes positively, even when there is a pragmatic demand to frame them negatively. Put together, these results shed light onto the question of how language users might strategically explore different linguistic strategies to communicate quantity in pragmatically favorable ways, and in particular how quantifiers are used vis-à-vis other lexical expressions of quantity.

1. Introduction

Talking about quantities is a routine activity which seems to pervade all spheres of human language use, from spontaneous conversation to highly technical and specialized communication. Natural languages themselves have a variety of ways of expressing quantity, from numeral systems (Comrie, 2022) to specialized quantity words like quantifiers (e.g., English *all*, *several*, *most*; Bach, Jelinek, Kratzer, & Partee, 2013;

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Matthewson, 2008) and other lexical expressions of quantity like collective nouns (e.g., English *the majority*). In the present study, we investigate how quantifiers are used strategically to construe situations in pragmatically favorable ways. As discussed in the next section, quantifiers have traditionally been studied in terms of their logical properties as well as in terms of how their interpretation in specific discourse contexts may or may not lead to the derivation of pragmatic inferences. However, while previous work has shed light onto the question of how quantifiers are used to communicate particular quantities, little is known about how their usage relates to that of other linguistic expressions which may serve as salient competitors in communication (Cummins, 2015), and how all of this is geared towards the realization of strategic goals.

In two experiments, we investigate how English speakers use quantifiers to present quantitative information in either a positive or negative light. In Experiment 1, we tap into people's biases when using a restricted range of four specific quantifiers, *all*, *most*, *some*, and *none*. In Experiment 2, we elicit descriptions of the same stimuli used in Experiment 1, annotating them for different types of quantity expressions and comparing them to the patterns found in Experiment 1. We analyze the data from both experiments in terms of a measure which captures the intuition that, in our design, certain stimuli are easier to describe positively without lying, while others are easier to describe negatively without lying. Crucially, we expect this difficulty to counteract any biases related to a straightforward informative usage of quantifiers, as would be expected under a general pragmatic account of quantifier usage according to which people produce the strongest possible statements they know hold true.

The results of both experiments show that people have clear preferences when using quantifiers to describe quantitative outcomes, and these preferences are adapted to whether the communicative context asks for positive or negative construal of the stated facts. We found that, as the difficulty of construing a particular piece of quantitative information positively or negatively increases, e.g., describing an objectively bad state of affairs as a good outcome, participants opted more frequently for informationally weaker expressions, i.e., quantifiers that are compatible with a larger range of plausible values such as *most* and *some*, as opposed to informationally stronger

quantifiers such as *all*. We also found that even when asked to frame quantitative facts negatively, participants preferred using positive rather than negative predicate statements (e.g., 'correct' as opposed to 'false'). Before reporting our results in more detail, we review previous work on the use and interpretation of quantifiers, and relate it to other work dealing with how quantity expressions can be used argumentatively.

1.1. Quantifier use and interpretation

Quantifiers have long been studied in semantics and pragmatics. A common framework for analyzing their meaning as well as the meaning of other linguistic expressions of quantity is the so-called generalized quantifier theory (Barwise & Cooper, 1981; see also Szymanik, 2016), which treats all expressions of quantity alike as establishing relations between sets, whether of individuals or degrees (see Solt, 2009, 2015). Generalized quantifier theory has provided a foundation upon which to explain pragmatic inferencing on the basis of expressions of quantity (Geurts, 2003), and plenty of empirical work has, for example, investigated the quantity-based implicatures that are related to the (under)informative usage of such expressions in context. One example is that of when *some* as in *Some of the students got the exam questions right* is taken to mean 'some but not all of the students' or 'some and potentially all students' as opposed to 'some and in fact all students' (see Breheny, 2019, and Reboul & Stateva, 2019, for recent overviews of work on such inferences).

All in all, this line of work has shown not only that language users have relatively fine-grained representations of the meanings of different quantifiers but also that they have expectations about which expressions might be used more appropriately in different contexts, even when their denotations partially overlap. While the bulk of empirical studies in this tradition have relied on acceptability and truth-value judgement tasks, some studies have, instead, focused on the production of quantifiers (see Moxey & Sanford, 2000, for an overview of related work). One recent study in particular has looked at the likelihood of different quantifiers being used to refer to different numerical ranges: van Tiel, Franke, and Sauerland (2020) asked English speakers to look at visual displays containing 432 circles colored in different proportions of black to red and to complete sentences of the sort "___ are red" by typing in their responses

into a text box. Their results suggest that people use quantifiers in a gradient way such that there is a lack of clear boundaries between the ranges captured by different expressions. In particular, van Tiel et al. (2020) found that, generally, non-endpoint quantifiers have broader ranges than the two endpoint quantifiers *none* and *all*, and that some non-endpoint quantifiers such as *some*, *several*, and *many* have particularly broad ranges compared to other non-endpoint quantifiers such as *a few* or *most*. Perhaps more interestingly, in their data the two endpoint quantifiers *none* and *all* have themselves quite broad ranges, where, for instance, *all* is occasionally used to refer to proportions around 3/4 of the circles, extending far out from the endpoint and into the upper range covered by *most*. This suggests much more gradience in quantifier usage than what would be expected under traditional accounts of quantifier meaning.

One way of interpreting these findings is to say that quantifiers with narrower ranges of values they map onto are more informative than quantifiers with broader ranges, such that they encode more precise numerical information. For instance, even though according to the van Tiel et al. (2020) data both *many* and *most* can be used to refer to proportions around 3/4 of the total set size, saying that *Most of the students in the class got a perfect score* encodes a more precise range of values than saying that *Many of the students in the class got a perfect score* by virtue of *most* being associated with a narrower range of values clustered around the value corresponding to 3/4 of the total set size. Crucially for our purposes, *most* can thus be said to be more informative or informationally stronger than *many*, given that, in the absence of other cues which might modulate their respective range of values, *most* encodes more precise information than *many*.

In the present study, we look at how informativity in the usage of quantifiers – roughly defined as the informational strength associated with particular quantifiers – varies as a function of strategic demands related to the communication of quantity. While it is generally assumed that language users strive to be as informative as possible, such that they pick the strongest possible linguistic forms when their goal is to communicate a given state of affairs as informatively as possible, this may not always be the case in actual communication. In particular, we explore how people's choice of quantifiers might be affected by the demand to argue for outcomes which may be at

odds with a truthful description of a given state of affairs.

1.2. Argumentative use of quantity expressions

Taken together, previous research on quantifiers shows that there is a great deal of variability in how they are interpreted and used. Particularly relevant for our purposes is the fact that language users adjust how they use and interpret such expressions on the basis of pragmatic expectations. The last couple of decades have seen a shift from looking primarily at the semantic properties of quantity expressions towards conceptualizing their usage as a process guided by argumentative and rhetorical goals. For instance, while similar in terms of the ranges of quantities they denote, *Most students passed the test* and *Not all students passed the test* might have quite different pragmatic effects on one's interlocutors just by virtue of foregrounding opposite polarities of the same event: *Most students passed the test* seems, a priori, quite neutral regarding the outcome, whereas *Not all students passed the test* might be said to imply that it was expected that all students would pass the test. Moxey and Sanford (2000) were among the first to note that, traditionally, research has looked at how linguistic expressions of quantity denote particular amounts or ranges of amounts instead of looking at "what different patterns of thought and reasoning might be set up by numbers and verbal expressions which might denote the same amount" (p. 239).

Based on the findings of some of their own empirical research on the interpretation of quantity expressions (e.g., Moxey & Sanford, 1993a, 1993b, 1997), Moxey and Sanford (2000) underline the fact that, even though there might be an overlap between the quantitative ranges different expressions are assigned to, people pick particular linguistic forms over others given that they have a perspective on the information they are conveying, one which not only affects their choice between competing forms but also the affective polarity of how they frame a particular outcome. These authors draw particular attention to the case of negative quantifiers, which include negated quantifiers as in *not many* but also quantifiers with a negative polarity like *no one*. Relevant to their argument is the fact that these quantifiers might induce different types of inferences compared to positive quantifiers (e.g., *many*, *a few*), such that utterances containing negative quantifiers are likely to induce interpreters to consider reasons as

for why the predicate might not hold, e.g., "In the aircrash, *not many* people were killed" might suggest that there is reason to suppose that *many* people might have been killed. Even quantifiers that have the same polarity and quite similar meanings can have different argumentative effects. For instance, Katzir and Ariel (2023) have shown that English *several* contributes a stronger argument than *a few*, which only contributes a weak or neutral argument, despite both denoting small quantities, as also shown by van Tiel et al. (2020).

Not only quantifiers may be used strategically to induce different interpretations in one's interlocutors. Indeed, Krifka (2005) argued that numbers too may induce different interpretations depending on whether or not they are rounded, such that a round number like *forty* (as opposed to *thirty-nine*) is interpreted in an approximate rather than in a precise way due to a pragmatic preference for simple over complex expressions. He related this preference for simplicity in form to a more general preference for communicating more cognitively salient representations, an argument which has found empirical support in a study by Solt, Cummins, and Palmović (2017). Krifka's general argument was that, in explaining pragmatic principles of usage, focus should be put not only on the relative simplicity or complexity of coding a particular message but also on the simplicity or complexity of what is communicated in the first place.

Insights such as the ones from Moxey and Sanford (2000) and Krifka (2005) have recently started gaining more traction in empirical research on quantity expressions. For instance, Cummins and Franke (2021) showed that language users produce utterances containing numerical expressions in ways which tend to enhance their argumentative strength, even if that comes at the price of potentially coming across as uncooperative. Using a corpus of university research rankings, they showed that in reporting the results of such evaluations university representatives tend to use the best available reference point, e.g., saying that their university is in the Top 20 even when they are actually the 18th top institution, and they tend to prioritize favorable outcomes over unfavorable ones, e.g., given two sub-rankings, mentioning only the one in which the institution fared better. These empirical results provide evidence that language users do not remain neutral when reporting quantitative information. More specifically, they provide evidence that language users systematically make efforts to

maximize the argumentative strength of utterances containing quantity expressions.

1.3. Current study

Building on the body of work just reviewed, the present study investigates how quantifiers may be used argumentatively when people report quantitative information. As mentioned earlier, we report two production studies focused on how English speakers modulate their usage of quantifiers when asked to describe the results of school exams either positively or negatively. In Experiment 1, a forced choice task, we instructed participants to fill in sentences describing stimuli that depict a range of different numerical facts. Participants had to choose between different expressions presented to them on a screen, including four different quantifiers and two adjectives. In Experiment 2, a free production task, participants came up with their own descriptions of the stimuli. By focusing on a restricted set of expressions whose meaning is theoretically well-understood, Experiment 1 aims to reveal usage patterns which can be reasonably said to relate primarily to the informational contrast between these different expressions. In contrast, by allowing participants to produce their own descriptions, thus eliciting a rather unconstrained set of quantity expressions, Experiment 2 aims to reveal usage patterns which provide a more comprehensive window into the actual production preferences involved in the communication of quantity. As discussed in more detail below, we analyze the data from both experiments in terms of a measure of argumentative difficulty which allows us to compare how, in our design, the production of utterances containing quantifiers is modulated by the difficulty associated with encoding particular states of affairs in maximally informative ways, both when there is competition only between quantifiers (Experiment 1) as well as when quantifiers are produced alongside other lexical expressions of quantity (Experiment 2).

Based on the findings discussed above we put forward some general conjectures regarding the usage patterns we might expect to see in our data. It is worth highlighting that while these are not stringent predictions which follow from particular hypotheses derived from the literature, they are empirically-grounded assumptions regarding the usage of quantity expressions which can be systematically teased apart, as we do in our study. Our first assumption relates to a general bias which might affect how language

users talk about the world, and by extension how our participants decide to frame their descriptions of quantitative outcomes. Several findings in the literature leads us to expect that even if asked to frame a situation negatively, participants may have a general bias towards positive framing. The "Pollyanna hypothesis" suggests that, in English, there are overall on average more positive than negative words, and positive words also get used more often (Dodds et al., 2014; Warriner & Kuperman, 2015). It is also the case that specifically when talking about quantities, English speakers generally focus on "increasing" or "added" quantities, as opposed to "decreasing" or "subtracted" ones (Winter, Fischer, Scheepers, & Myachykov, 2023). We also know from the literature on gain-loss framing that framing the same outcome as gain rather than loss induces positive emotions (Nabi et al., 2020), and framing a health procedure in terms of survival as opposed to mortality can lead to higher adoption (Armstrong, Schwartz, Fitzgerald, Putt, & Ubel, 2002). Thus, even if we ask participants to construe a given state of affairs as negatively as possible, they may still be affected by biases leading them to choose a positive framing as the default option. In Experiment 1, we take participants' choice of adjective as indicative of their framing preferences. In Experiment 2, we extend our analysis to other polarity markers, including adverbs (e.g. *correctly*, *badly*) and verbs (e.g. *ace*, *fail*).

Then, similarly to what has been reported by Cummins and Franke (2021), we assume that in our data too people might have a general preference for focusing on favorable outcomes relative to unfavorable ones, at least if a favorable outcome is defined as one which is pragmatically favorable to a producer given their subjective goals. Crucially, however, if a producer's goal is to make a situation appear bad, then framing it as a negative outcome is favorable to them. Of course, favorable outcomes may also be defined as outcomes which are *objectively* positive: a high rate of correct answers on an exam is objectively positive but *subjectively* negative if one's goal is to frame the situation as a bad outcome. In their analysis of utterances containing numerical expressions, Cummins and Franke (2021) treated producers prioritizing favorable outcomes to mean that they have a preference to frame outcomes in ways which are favorable to them given their own argumentative goals. Crucially, people were shown to frame outcomes in potentially objectively less favorable ways as long as the framing

was in line with their subjective agenda. In our analysis, we treat this assumption the same way.

This leads us to our last assumption, which relates to the difficulty associated with encoding objectively good or bad outcomes in pragmatically favorable ways. Indeed, it may be that, as the difficulty in describing a situation in a maximally informative way increases, so does a producer's tendency to use particular expressions, or combinations of particular expressions. One key factor which is expected to constrain people's choice of different linguistic expressions is that between being truthful to the data they are describing, on the one hand, while remaining faithful to their own argumentative goals, on the other. One rather straightforward way of potentially explaining variability in how participants in our task choose to express quantities is to explore how different linguistic preferences vary depending on how easy it is to describe particular stimuli either as a good or as a bad outcome, which we capture in a gradient variable that we call *argumentative difficulty*, as discussed in more detail in section 2.2.2. In particular, we use this measure to analyze participants' usage of quantifiers in both Experiment 1 and 2, drawing on the notion discussed in section 1.1. that different quantifiers have different informational strengths.

2. Experiment 1 - Forced choice task

2.1. Method

2.1.1. Participants

Sixty self-reported native speakers of English were recruited via the Prolific crowdsourcing platform. Participants were paid £2.50 for their participation in the experiment.

2.1.2. Design and procedure

The task involved describing visual stimuli consisting of tables representing the outcome of a school exam, as shown in Figure 1. Participants were instructed to describe the stimuli in terms of either a high or low success rate, without resorting to false de-

scriptions, that is, without outright lying. Each participant saw, in random order, one of the 20 tables shown in Figure 2 below, where, for instance, the table at the bottom right is a maximally positive outcome whereas the table in row 2, rightmost column, is a maximally negative outcome. Half of the stimuli were presented in a high success condition, where participants were prompted to describe the stimuli as if the students had done well (see Fig. 1); the other half were presented in a low success condition, where participants were prompted to describe the stimuli as if the students had done poorly. The allocation of stimuli to high vs. low success conditions was randomized.

Participants were presented with the sentence frame "In this exam ___ of the students got ___ of the questions ___". The first two gaps could be filled with one of four different quantifiers from the set *all*, *most*, *some*, *none* while the last gap could be filled with one of two adjectives, *right* and *wrong*. Before the test trials, participants were shown practice trials which mimicked the exact procedure of the test trials. As discussed in the introduction, the contrast between *some* and another quantifier with a partially overlapping meaning such as *most* is empirically interesting as *some* may be used with the explicit intent of rendering a description underinformative. For instance, in the high success condition, one could describe the table at the bottom right of Fig. 2 (a maximally positive outcome) with "In this exam *all* of the students got *all* of the questions *right*". The same table in the low success condition instead is more likely to be described with a weaker statement, such as "In this exam *some* of the students got *some* of the questions *right*". Similarly, in the high success condition, one could describe the table in row 2, rightmost column, of Fig. 2 (a maximally negative outcome) with "In this exam *some* of the students got *some* of the questions *wrong*". The same table in the low success condition instead is more likely to be described with a stronger statement, such as "In this exam *all* of the students got *all* of the questions *wrong*".

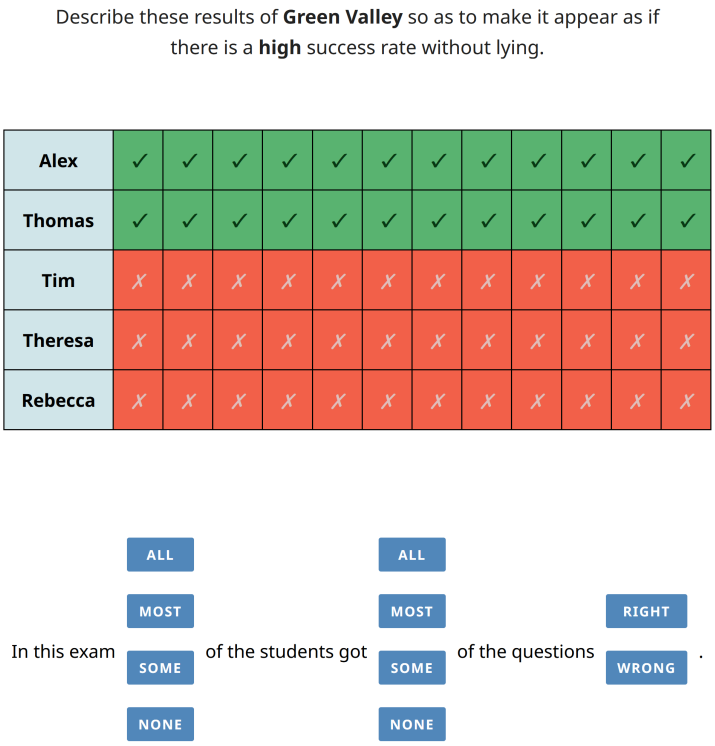

Figure 1.: Example trial. Top shows an example situation in a particular experimental condition (varying across trials); bottom shows response options available to participants (constant across trials).

2.1.3. Materials

Participants saw 20 different tables, each comprised of five rows, each row representing one student, and twelve columns, each column representing one exam question, as shown in Figure 2. The students were identified by unique names (e.g., Alex, Thomas, Theresa, etc.), and each question was marked as having been responded to either correctly – represented by a check mark against a green background – or incorrectly – represented by an X against a red background, as shown in Fig. 1 above.

2.1.4. Statistical analyses

We analyzed the data using R version 4.2.1 (R Core Team, 2014) and the packages *plyr* version 1.8.7 (Wickham, 2011), *ggtext* version 0.1.2, *ggalluvial* version 0.12.5 (Brunson & Read, n.d.), *brms* version 2.18.0 (Bürkner, 2017), *tidybayes* version 3.0.2 (Kay,

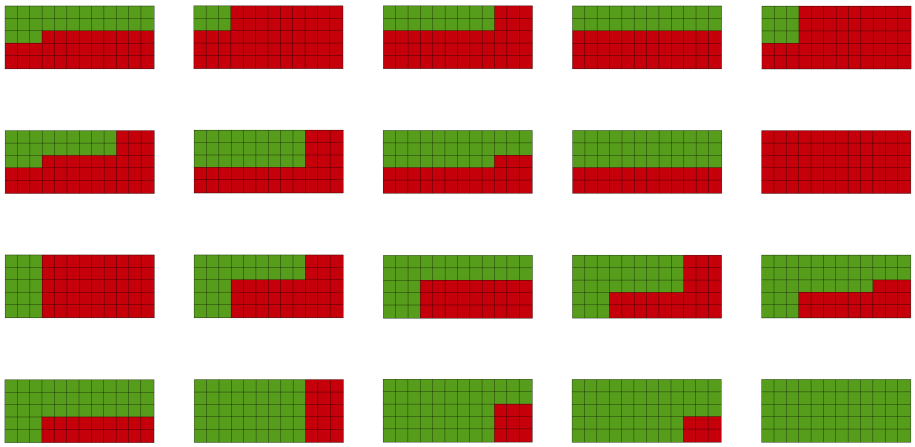

Figure 2.: Visual stimuli. The ratio of green to red cells in each table indicates the rate of success at that particular exam: a higher ratio of green cells indicates a better relative outcome.

2023), and *posterior* version 1.3.0 (Bürkner, Gabry, Kay, & Vehtari, 2023), in addition to the *tidyverse* version 2.0.0 (Wickham et al., 2019)[1]. We fitted Bayesian linear mixed effect models separately for each choice (each gap in the example sentence). For the choice of adjective, this was a logistic mixed model with fixed effect 'condition' (**Low** vs. **High** condition, **Low** = reference level). For the choice of quantifier, the models were mixed multinomial models with fixed effects 'adjective' (*wrong* vs. *right*, *wrong* = reference level) and 'condition' (**Low** vs. **High**, *Low* = reference level), and the interaction between the two. The adjective model estimated the log odds of each observation being the adjective *right* as opposed to *wrong*, while the quantifier models estimated the log odds of each observation being either the quantifier *none*, *some*, or *most*, respectively, as opposed to *all*. The adjective model included random intercepts for participants and items, as well as by-participant and by-item random slopes for the condition effect. The quantifier models included the same random effects, and on top of it also by-participant and by-item random slopes for the adjective effect. All models were fitted using the default priors from *brms* (Bürkner, 2017) and had the following forms, shown in *brms* syntax:

[1]All materials, data, and analysis scripts can be found in the following repository: `https://tinyurl.com/yrasxa9r`

Adjective model (logistic)	**Quantifier models (multinomial)**
adjective ∼ condition +	quantifier ∼ adjective * condition +
(1 + condition | participant) +	(1 + adjective + condition | participant) +
(1 + condition | item)	(1 + adjective + condition | item)
family = bernoulli()	family = categorical()

2.2. Results

2.2.1. Descriptive results

Figure 3 shows the different combinations of quantifiers and adjectives attested in the data, split over the two experimental conditions **High** and **Low**. The most obvious difference between conditions is that participants demonstrated a clear preference to choose the adjective as a function of condition: in the **High** condition they chose the adjective *right* in 92% of cases (β = .99, 95% CrI [0.96, 1.0]), as opposed to only 18% in the **Low** condition (β = .10, 95% CrI [0.05, 0.17]).

In both conditions, *some* and *most* are the quantifiers most frequently used to refer to students (e.g., In this exam *some/most* of the students ..., 38% and 40%, respectively, in the **High** condition and 38% and 36%, respectively, in the **Low** condition). This usually happened in combination with *most* and *all* referring to questions (i.e., In this exam *some* of the students got *most/all* of the questions ..., 41% and 36%, respectively, in the **High** condition and 39% and 33%, respectively, in the **Low** condition). Out of the four available quantifiers, *none* is used, in absolute terms, rather infrequently (59 times when referring to students and 29 when referring to questions). When used to refer to students, however, it occurs about twice as frequently in combination with the adjective of the opposite polarity of the respective condition (e.g., "In this exam *none* of the students got *all* of the questions **wrong**" in the **High** condition) than in combination with the adjective which matches the polarity of the condition (e.g., "In this exam *none* of the students got *all* of the questions **right**" in the **High** condition).

Figure 4 shows the estimated log odds for each pairwise comparison between *all*

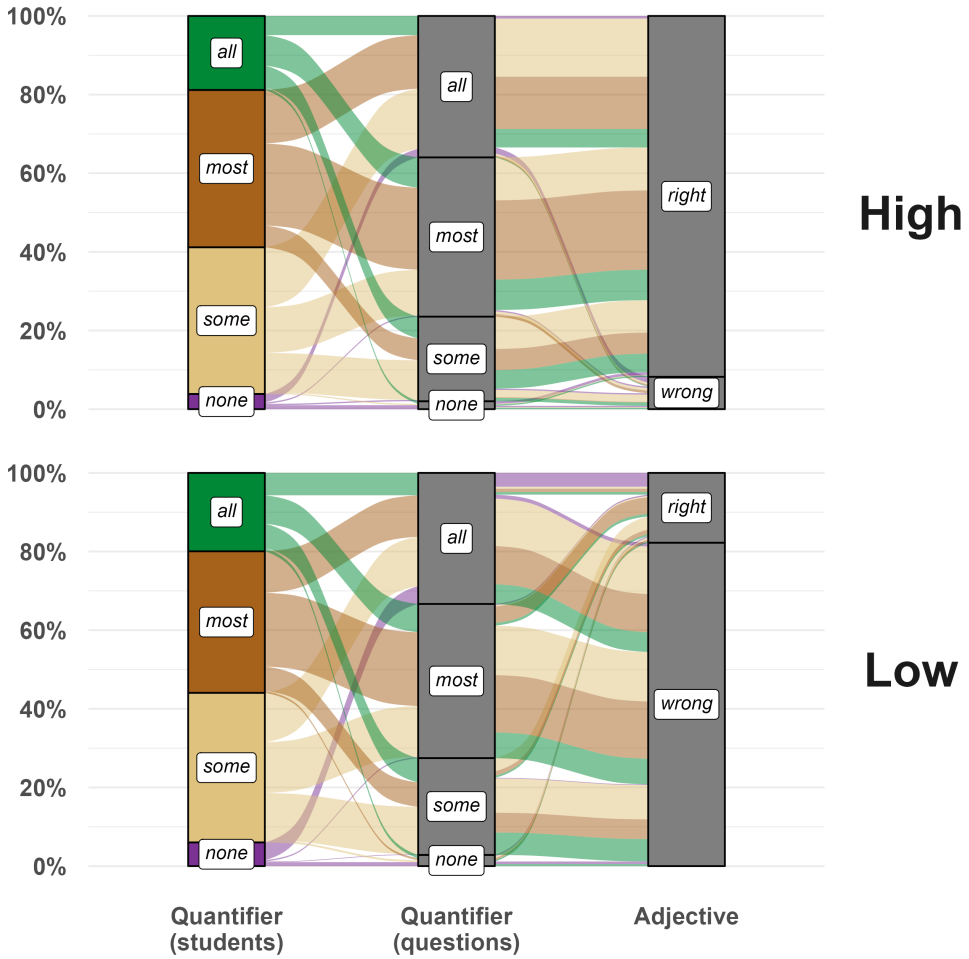

Figure 3.: Relative frequency of each expression attested in the data, shown in terms of the two experimental conditions.

and one of the other three quantifiers for each adjective-condition pair when referring to students (left) and questions (right), as retrieved from the posterior samples of the quantifier models. While the estimates provide support to the empirical findings reported above, the figure also suggests that there is a great deal of uncertainty around the estimates, in particular when it comes to *none*. Ultimately, the uncertainty seen in the figure suggests that the experimental condition together with the choice of adjective account for only part of the variation seen in terms of the quantifier choice.

2.2.2. Argumentative difficulty

In the previous section, we presented an overview of the preferences participants showed when describing the stimuli. Of course, such absolute preferences speak lit-

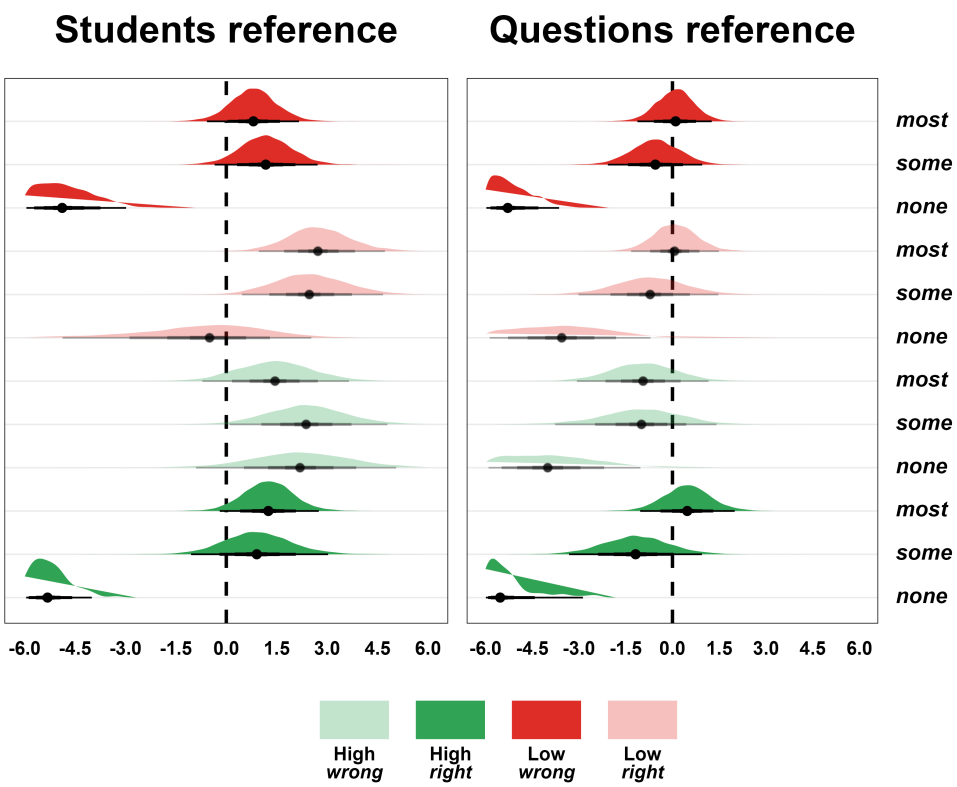

Figure 4.: Differences in the (posterior) log odds of *none*, *some*, and *most* being used to refer to students (left) and questions (right) relative to *all* for each adjective-condition pair. The more positive the value associated with a given quantifier on the right margin of the plot, the higher the odds of that quantifier occurring compared to *all* for the respective adjective-condition pair.

tle about variation which might be associated with the strategic usage of quantifiers in situations in which people have to describe outcomes which are, objectively, not in line with the argumentative goals represented by the **High** and **Low** experimental conditions. A straightforward way of measuring such argumentative difficulty would be to match the proportion of correct answers against the experimental conditions, where, for instance, a proportion of 0 would receive a minimum rating of 0.0 in the **Low** condition and a maximum rating of 1.0 in the **High** condition. In other words, a rating of 0.0 would index the easiest possible situation in a given condition, while a rating of 1.0 would index the most difficult. To derive ratings for the other (non-extreme) situations in the **High** condition, one can subtract the value from the maximum rating of 1.0. Thus, for instance, a stimulus depicting 15 correct answers and 45 incorrect answers (as in Fig. 7) receives a rating of 15/ 60 = 0.25 in the **Low** condition and

1 - (15/ 60) = 0.75 in the **High** condition. We use an approach where we also take into account features of the distribution of correct/ incorrect answers, in particular the number of students who answered zero questions correctly[2]. For instance, for any proportion of correct answers, we divide that number by the total number of students scoring exactly zero, in the case of the **High** condition also subtracting that value from the maximum rating of 1.0, e.g., 15 correct answers, where 3 students answered 5 questions correctly each and 2 students answered 0 questions correctly = 1 - ((15/ 60)/ 2) = a rating of 0.875.

Figure 5 shows the probability of each quantifier being used to refer to students as a function of the difficulty metric just discussed. The figure shows that, when referring to students in the **High** condition and framing their description using the adjective *right*, participants were more likely to use *all* at around 0.0 difficulty. From around 0.0 difficulty there was a steady increase in the probability of *most* being used, to the detriment of *all*. At around 0.5 difficulty, *some* was the most likely choice. From around 0.75 difficulty, *none* became the most likely choice. When referring to the students in the **High** condition and framing their description using the adjective *wrong*, participants were more likely to use *none* from 0.0 to around 0.75 difficulty.

When referring to students in the **Low** condition and framing their description using the adjective *wrong*, participants were more likely to use *all* at around 0.0 difficulty. From around 0.0 difficulty there was a steady increase in the probability of *most* being used, to the detriment of *all*. From around 0.35 difficulty participants only used *some*. When referring to students in the **Low** condition and framing their description using the adjective *right*, using *most* was the most likely choice at the [0.0, 0.5] interval. From stimuli with at least 0.5 difficulty, *all* became the most likely choice.

Figure 6 shows the probability of each quantifier being used to refer to questions as a function of the argumentative difficulty. The figure shows that, when referring to students in the **High** condition and framing their description using the adjective *right*, participants were more likely to use *all* at around 0.0 difficulty. From around 0.0 difficulty there was a steady increase in the probability of *most* being used, to the detriment of *all*. At around 0.5 difficulty, all quantifiers except for *none* seem

[2]Note that the results would look qualitatively similar if one were not to take the number of students who scored zero in the exam into account, despite this being a more balanced metric.

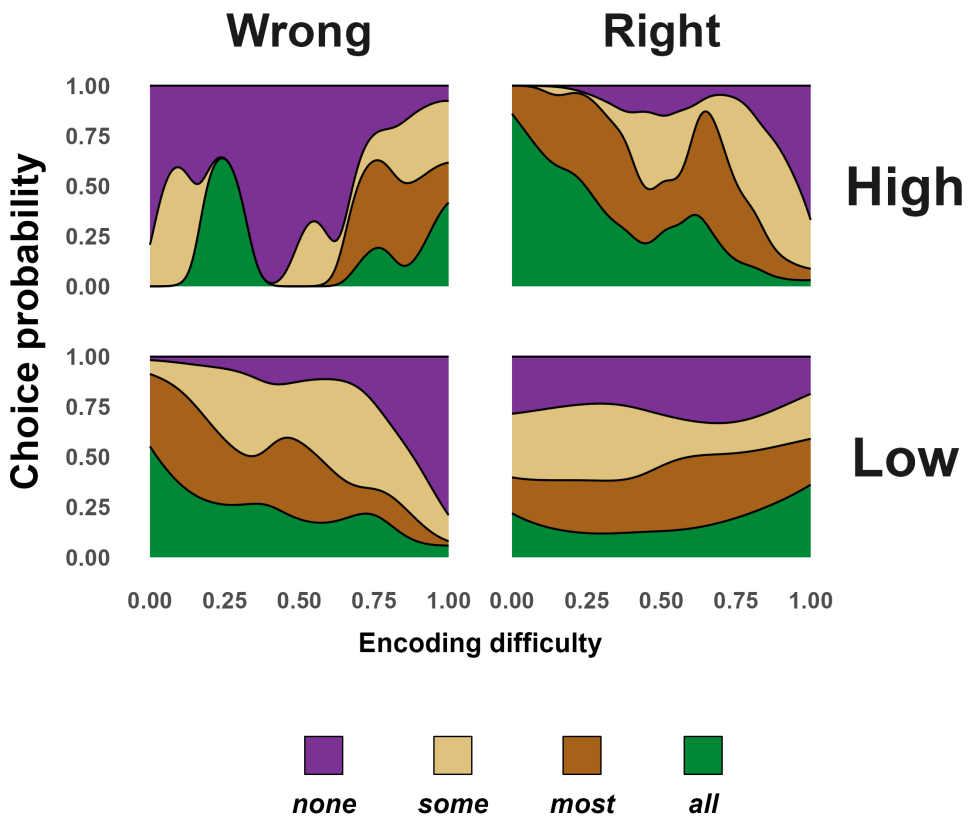

Figure 5.: Probability of *all*, *most*, *some*, or *none* being used to refer to students in Experiment 1 as a function of argumentative difficulty, shown in terms of the experimental conditions (**High** vs. **Low**) and adjective (**Right** vs. **Wrong**). The x-axis indicates the varying degrees of difficulty whereas the y-axis indicates the probability of any given quantifier being used.

about equally likely to be used. From around 0.75 difficulty, *none* became the most likely choice. When referring to the students in the **High** condition and framing their description using the adjective *wrong*, participants were more likely to use either *all* or *some* at around 0.0 difficulty. At around 0.5 difficulty, *all* and *most* are the most likely choices. *none* is the most likely choice both at around 0.75 and 1.0 difficulty.

When referring to students in the **Low** condition and framing their description using the adjective *wrong*, participants were more likely to use *all* at around 0.0 difficulty. From around 0.0 difficulty there was a steady increase in the probability of *most* being used, to the detriment of *all*. From around 0.75 difficulty *some* became the most likely choice. When referring to students in the **Low** condition and framing their description using the adjective *right*, participants were more likely to use *none*

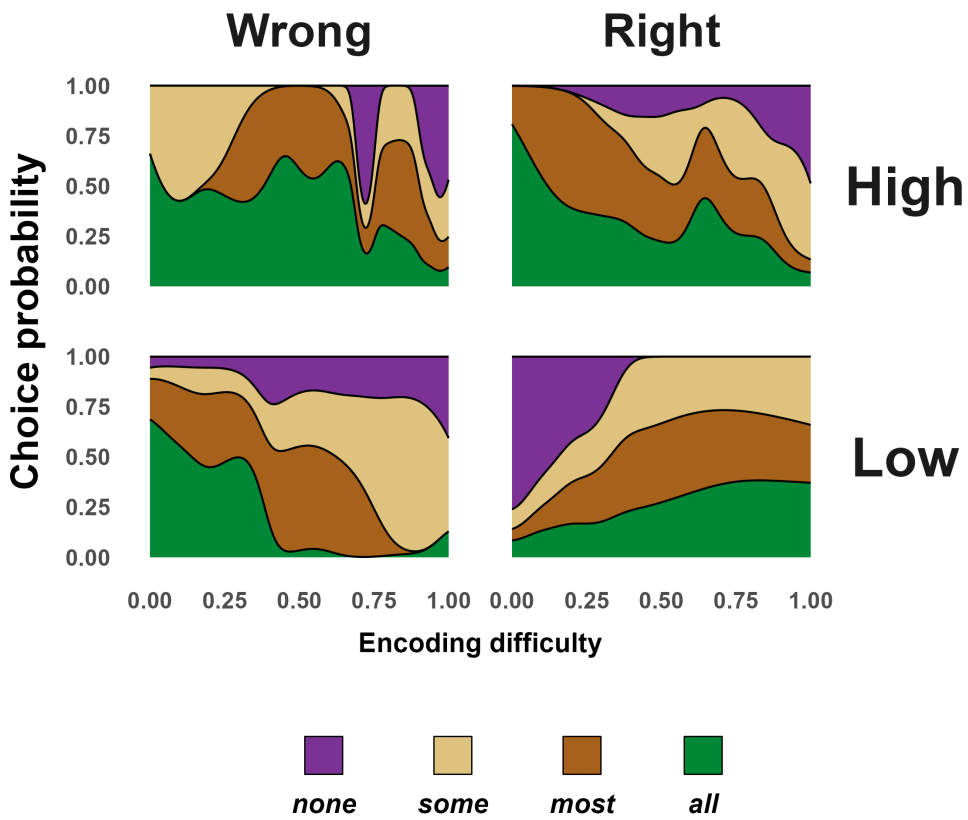

Figure 6.: Probability of *all*, *most*, *some*, or *none* being used to refer to questions in Experiment 1 as a function of argumentative difficulty, shown in terms of the experimental conditions (**High** vs. **Low**) and adjective (**Right** vs. **Wrong**). The x-axis indicates the varying degrees of difficulty whereas the y-axis indicates the probability of any given quantifier being used.

at around 0.0 difficulty. From around 0.0 difficulty there was a steady increase in the probability of all other quantifiers being used, to the detriment of *none*.

2.3. Discussion

The descriptive results show that people choose to frame their descriptions virtually always in line with the experimental condition, that is, as a positive outcome (i.e., *right*) in the **High** condition and as a negative outcome (i.e., *wrong*) in the **Low** condition. As for people's choice of quantifiers, the results show that they have a preference for using *some* and *most* to refer to students (e.g., "In this exam *some*/*most* of the students ..."), usually in combination with *most* and *all* referring to questions (e.g., "In this exam *some* of the students got *most*/*all* of the questions ..."). We also

addressed the issue of how the difficulty in describing particular stimuli in either the **High** or **Low** experimental condition affects people's choices of quantifiers. We found that people tended to use informationally weaker quantifiers as the argumentative difficulty increases, at least when framing the exam outcomes in line with the respective experimental conditions. In other words, people are more likely to shift from using *all* towards *most* and *some* as argumentative difficulty increases. However, this pattern only characterized those situations where the framing chosen by the participants matched that of the condition (e.g., talking about 'right' answers in the **High** condition). When this was not the case, results were rather unsystematic, with a fairly similar response distribution regardless of argumentative difficulty in the **Low** condition, and an overall tendency to pick *none* in the **High** condition. In Experiment 2, a free production task, we tap into people's biases when being able to construe the outcomes without being constrained by a specific set of quantifiers and adjectives.

3. Experiment 2 - Free production

3.1. Method

3.1.1. Participants

Thirty self-reported native speakers of English were recruited via the Prolific crowdsourcing platform. Participants were paid £2.50 for their participation in the experiment.

3.1.2. Materials, design and procedure

Participants were instructed to describe a random subset of 12 of the stimuli used in Experiment 1. However, instead of being asked to fill gaps in a sentence, they were presented with a text box, as shown at the bottom of Figure 7, and instructed to type their own descriptions of the stimulus into the box. To ensure that responses were not too short, the typed text had to have at least 11 characters before a "next" button allowed participants to proceed to the next trial appeared on screen. Before the test trials, participants were shown a sample stimulus and given two example descriptions,

Most of the students got most of the questions right as an example of a description framing the results in terms of a high success outcome and *Only a few students got a full score* as an example of a description framing the results in terms of a low success outcome. The provided examples included quantifiers so as to prompt participants to produce (more) descriptions containing quantifiers and other quantity expressions as opposed to, for example, vague adjectives (e.g., *The students did really well*) or other linguistic strategies (e.g., *The students aced the exam*).

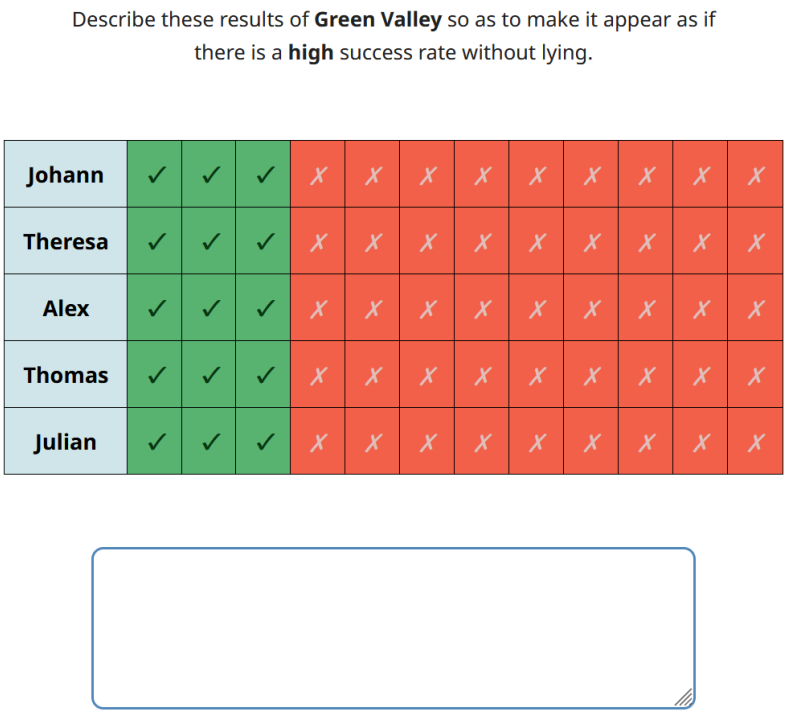

Figure 7.: Example trial.

3.1.3. Data annotation

Participants produced descriptions of various lengths and degrees of complexity, as shown in examples (1)- (5) below. The average number of words per response was 11 words long, the shortest description being 3 words long (1) and the longest 35 words long (2).

(1) *all students failed*

(2) *Green Valley will be able to achieve 100% pass rate on testing and this can be 2 out of 5 students and at least 3 out of 5 students will gain a 60% pass rate*

Given that we were primarily interested in participants' usage of quantity expressions, and even more specifically in their usage of quantifiers, we developed an annotation system geared towards capturing those responses that resembled the descriptions from Experiment 1, in other words, utterances referring to either concrete numbers or proportions of students and/ or exam questions. Out of the 330 recorded descriptions, 81% of them consisted of simple sentences containing a single reference to students and/ or questions, as in (1) above and (3) below. The remaining 19% of the data consisted of more complex descriptions containing multiple references to students and/ or questions, either separated by a comma, period, or a semicolon, as in (4), or connected by a conjunction like "and" or "but", as in (2).

(3) *Majority of our students passed with full marks*

(4) *Nobody achieved a 100% score, the highest mark was 75%*

For comparability with Experiment 1, we decided to focus our attention on descriptions consisting of a single sentence, where any description containing multiple student/ question references was not included. 64 descriptions (19% of the original data) were removed from the main analyses but are discussed qualitatively at the end of section 3.2.1. Note that there were also three instances of complex constructions containing multiple quantity expressions within a single student/ question reference, as in (5). These were also removed from the analyses.

(5) *Scores were all 25% or below*

Expressions referring to the 'students' included *student(s)*, *pupils*, *child(ren)*, *the group*, *the people*, *no one*, *nobody*, and *everyone*. Expressions referring to the 'questions' included *(a) score(s)*, *grades*, *marks*, *answers*, *questions*, *responses*, and *success (rate)*. Some of these expressions served as self-standing referring expressions, as in (6), while others acted as modifiers of other expressions, as in (7) and (8).

(6) *<u>The majority</u> of the tests were good*

(7) *Most students only have a 25% success rate*

(8) *2 thirds of students had some success in the tests*

Given that over half of the descriptions contain two quantity expressions (55% of the data), one referring to the students (e.g., "Over half of pupils received a score of 75%") and one referring to the questions (e.g., "Over half of pupils received a score of 75%"), we decided to code each expression separately. We coded whether or not the expression referring to students was negated (e.g., *At Riverside, not all students scored perfectly*), and whether or not the predicate of the sentence was negated, regardless of what it referred to (e.g., *Most of the students didn't get all the questions right*). Each description was thus coded in terms of:

- Type of quantity expression referring to students - Whether the expression consisted of a numeral (e.g., *50%*), a quantifier (e.g., *some*), a nominal expression of quantity (e.g., *the majority*), or no quantity expression at all;
- Type of quantity expression referring to questions - Whether the expression consisted of a numeral (e.g., *50%*), a quantifier (e.g., *some*), a nominal expression of quantity (e.g., *the majority*), or no quantity expression at all;
- Negation student expression - Whether or not the expression referring to students was negated (e.g., *At Riverside, not all students scored perfectly*);
- Negation predicate - Whether or not the predicate was negated (e.g., *Most of the students didn't get all the questions right*);
- Argumentative polarity of the predicate - Whether the predicate was framed in terms of a positive, neutral, or negative outcome (e.g., *Majority of our students got answers correct*).

Following the coding scheme as just described, a description such as "Over half of pupils received a score of 75%" would be coded as in Table 1 below.

When it comes to annotating the argumentative polarity of the predicate, other than relying on clear linguistic markers of polarity such as adjectives/ adverbs (e.g., *correct(ly)*; *low*) and verbs (e.g., *aced*, *failed*), we focused on participants' usage of numerals. For the purposes of the annotation, in the absence of other polarity markers, we considered any numeral referring to a proportion below 0.5 (e.g., *25%*; *a third*; *zero*)

Expression type$_{\text{students}}$	Quantifier
Expression type$_{\text{questions}}$	Numeral
Students negation	No negation
Predicate negation	No negation
Predicate polarity	Positive

Table 1.: Sample coded description.

to encode a negative polarity. Conversely, we considered any numeral referring to a proportion above 0.5 to encode a positive polarity (e.g., *75%*; *9/ 12*; *three quarter*). In the absence of other polarity markers, a proportion of exactly 0.5 would be coded as neutral, but this was unattested for single-reference descriptions.

3.1.4. Interrater reliability

A second coder was given 50 responses randomly selected from the data set (17% of the original data). Both the percentage of reliability between both coders (a) as well as the Cohen's kappa (b) were calculated, as shown in Table 2.

Expression type$_{\text{students}}$	90%
Expression type$_{\text{questions}}$	94%
Students negation	94%
Predicate negation	92%
Predicate polarity	94%

(a) Percentage.

Expression type$_{\text{students}}$	0.85
Expression type$_{\text{questions}}$	0.83
Students negation	0.91
Predicate negation	0.76
Predicate polarity	0.89

(b) Cohen's kappa.

Table 2.: Reliability measures.

3.2. Results

3.2.1. Descriptive results

After removing any responses as described in section 3.1.3 above, there were 265 responses left. 55% of the descriptions contained both a quantity expression referring to the students and one referring to the questions, as in (7) and (8) above. 33% of the

descriptions, while containing a quantity expression referring to the students, involved the usage of vague predicates not containing a quantity expression, such as in (9) below. 9% of the descriptions did not contain any quantity expression, as in (10) and (11). Finally, the remaining 3% of the data, while containing a quantity expression referring to the questions, made no explicit reference to any students, such as in (12).

(9) *Two of our students performed well*

(10) *According to the answer table Julian, Julia and Tim have done well*

(11) *The results of the tests at Green Valley are challenging and student aptitude and lots of revision and hard work could achieve an optimum score and pass rate*

(12) *The success rate is more than 75% at this school*

Strategies involving quantity expressions

Before describing the overall responses, as aggregated over all stimuli, we look at the responses to the extreme cases, namely the all green and all red stimuli, which represent scenarios in which the exam was either a complete success or a complete failure, respectively. These extreme cases serve as a baseline against which to compare all other responses, as these constitute situations where, depending on the experimental condition, the pressure for participants to describe the stimuli in non-truthful ways is either marginal or as high as it can be. As there are only a few responses for each stimulus-condition pair, we restrict ourselves to a qualitative look at the data.

Looking at how the stimulus where all cells are green was described in the **High** condition (5 responses), we see that participants tend to produce strong statements containing the quantifier *all*, such as in (13) and (14) below. These tend to contain positive polarity markers such as *well* and *100%*.

(13) *According to the answer table All students have done well*

(14) *ALL students achieved 100%*

Looking at how the stimulus where all cells are red was described in the **Low** condition (8 responses), we see that participants tend to produce strong statements either containing the quantifier *all*, such as in (15) below, or a negative quantifier like *no*, as in (16) below. These tend to contain a negative polarity marker such as *bad*

when a positive quantifier is used, and a positive polarity marker such as *right* when a negative quantifier is used.

(15) *According to the table all have done very bad*

(16) *No-one got anything right*

Looking at how the stimulus where all cells are green was described in the **Low** condition (3 responses), we see that participants tend to produce strong statements containing the quantifier *all*, such as in (13) and (14) below. These tend to contain positive polarity markers such as *well* and *pass*.

(17) *All students performed well*

(18) *all students passed this exam, with the same score*

Looking at how the stimulus where all cells are red was described in the **High** condition (4 responses), we see that participants tend to produce strong statements containing the quantifier *all*, such as in (19) and (20) below. These utterances tend to contain predicates which are vague as to whether or not the class did well in the exam, such as *to be challenging* and *try their hardest*.

(19) *the exam was unusually challenging for all students*

(20) *All of the students tried their hardest*

In summary, when describing the extreme cases, participants tend to produce statements containing strong expressions such as the quantifiers *all* and *no*. In the **High** condition, when the stimulus was easy to describe the quantifier *all* tends to be paired with positive polarity markers such as 'well' and '100%'. In the **Low** condition, on the other hand, when the stimulus was easy to describe either *all* tends to be paired with negative polarity markers such as 'bad' or a negative quantifier tends to be paired with a positive polarity marker such as 'right'. In the cases where it was hard to describe the stimulus, participants more often than not produced utterances containing neutral predicates. As discussed in more detail in section X, we removed responses with a neutral predicate from the main analyses reported below.

Figure 8 shows the different combinations of quantity expressions attested in the data, aggregated over all stimuli. The figure shows the responses in terms of the exper-

imental conditions (**High** and **Low**) and in terms of how participants chose to frame their descriptions (**Positive** or **Negative** predicate polarities). Negated expressions are shown as self-standing categories. Darker and larger circles indicate a higher relative frequency of any given attested combination, the absence of a circle indicating the absence of any attested data point for a given combination.

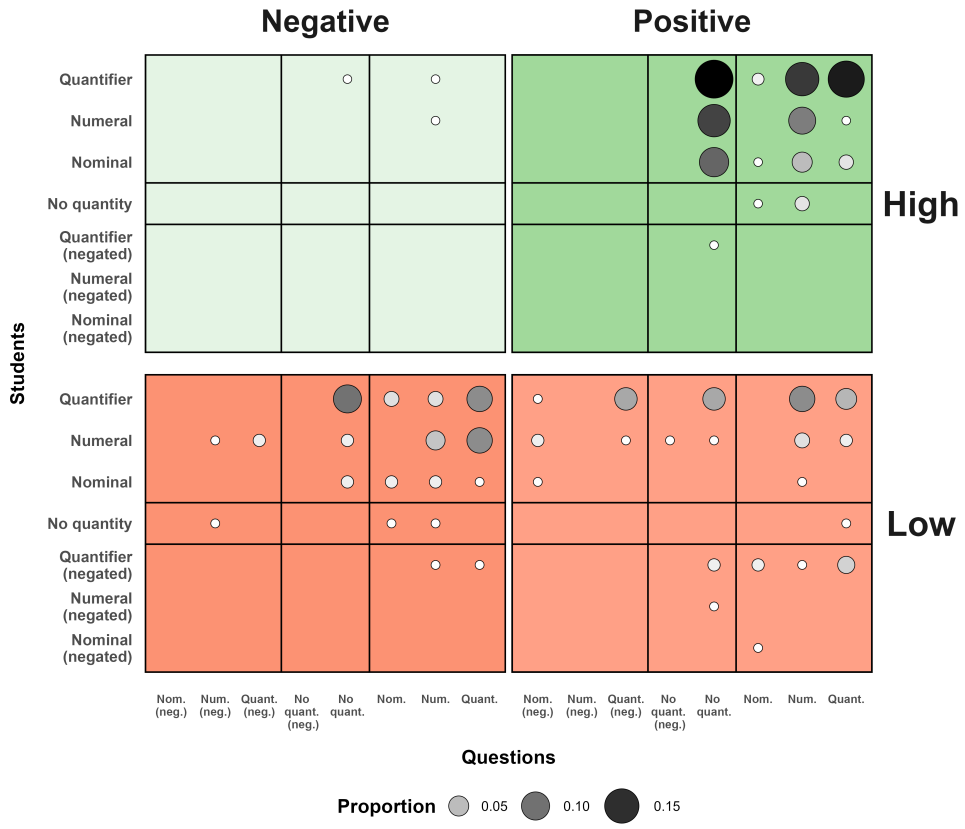

Figure 8.: Proportion of each type of quantity expression attested in the data relative to the experimental condition (**High** vs. **Low**) and predicate polarity (**Positive** vs. **Negative**). The x-axis indicates expressions which refer to the questions whereas the y-axis indicates expressions which refer to the students. Darker and larger circles indicate a higher relative frequency.

In the **High** condition (green tables, top row) 98% of the descriptions were framed positively, e.g., "All students got at least *75% correct*", whereas in the **Low** condition (red tables, bottom row) 51% of the descriptions were framed negatively, e.g., "over half of students got *0%*". In other words, while in the **High** condition there was a clear bias for framing descriptions positively, in the **Low** condition there was no particular framing bias. So, overall, there appears to be a bias for framing outcomes positively

(74% across conditions). Interestingly, in the **Low** condition, people used a wider range of strategies when framing their descriptions positively compared to when framing their descriptions negatively, as evidenced by the large number of circles plotted within each of the two red tables in the bottom row of Fig. 8.

The most prevalent strategies across conditions (20% of the responses) was to use quantifiers to refer to both the students and the questions (e.g., "*most* of the students answered *most* of the questions correctly"), closely followed by only using a quantifier to refer to the students without any quantity expression referring to the questions (19%, e.g., "*Most* of the students got full marks"). As evidenced by the large dark circles plotted in Fig. 8, in the **High** condition the most frequent strategy (20%) was to use a quantifier to refer to the students without any quantity expression referring to the questions. The second most frequent strategy (17%) consisted of using quantifiers to refer to both the students and the questions. In the **Low** condition, this was the most frequent strategy (24%). The second most frequent strategy in the **Low** condition (18%) consisted of using a quantifier to refer to the students without any quantity expression referring to the questions. Using a quantifier to refer to the questions without any quantity expression referring to the students is the least prevalent strategy both across conditions (less than 0.05%) as well as within each condition (no attested case in the **High** condition and less than 0.01% in the **Low** condition). When people frame their descriptions positively in the **Low** condition, they tend to use negated expressions quite frequently (20%), both in the form of negated references to students, as in (21) and (22) below, as well as in the form of negated references to questions, as in (23) and (24) below.

(21) *No* students got all the answers correct

(22) *Not all* students answered all of the questions correctly

(23) Some students *didn't get all* the questions right

(24) Over half of the students got *no* correct answers

Strategies involving quantifiers

Figure 9 zooms in on the descriptions attested in the data containing at least one quantifier (151 observations in total), shown in terms of the experimental conditions (**High** and **Low**) and in terms of how participants chose to frame their descriptions (**Positive** or **Negative** predicate polarities). To compare these results to Experiment 1, we highlight responses containing either *all*, *most*, *some*, and *none* and collapse all other quantifiers together. As the figure shows, the distributions follow the same general patterns as in the data containing all expressions (Fig. 8), where in the **High** condition there is a clear bias for framing descriptions positively (98%) while in the **Low** condition there is practically no bias for framing descriptions either positively (52%) or negatively (48%). This suggests a general bias for framing descriptions in terms of positive outcomes, even in situations where participants' task was to frame the outcome in terms of a low success rate.

Using a quantifier other than *all*, *most*, *some*, or *none* to refer to the students without any quantity expression referring to the questions (e.g., "Over *half* students got correct answers") is the most prevalent strategy across conditions (10%), followed by using *all* to refer to the students without any quantity expression referring to the questions (e.g., "*All* students show an average to high success pattern"; 9%). As evidenced by the large dark circles plotted in Fig. 9, in the **High** condition the most prevalent strategy (16%) is using a quantifier other than *all*, *most*, *some*, or *none* to refer to the students without any quantity expression referring to the questions (e.g., "*Several* of the students scored full marks"). The second most frequent strategy (15%) consists of using *all* to refer to the students with a numeral referring to the questions (e.g., "*All* students had more than *75%* correct"). In the **Low** condition the most prevalent strategy (16%) is using a numeral to refer to the students with a quantifier other than *all*, *most*, *some*, or *none* referring to the questions (e.g., "*40%* of students failed *every* question"). The second most frequent strategy (8%) consists of combining two quantifiers other than *all*, *most*, *some*, or *none* (e.g., "*A few* in the group got *every* question wrong"). The usage of *all*, *most*, *some*, and *none* corresponds to 67% of all attested quantifier usages (101 observations in total), utterances with either *all* or *most* making up 54% of the data.

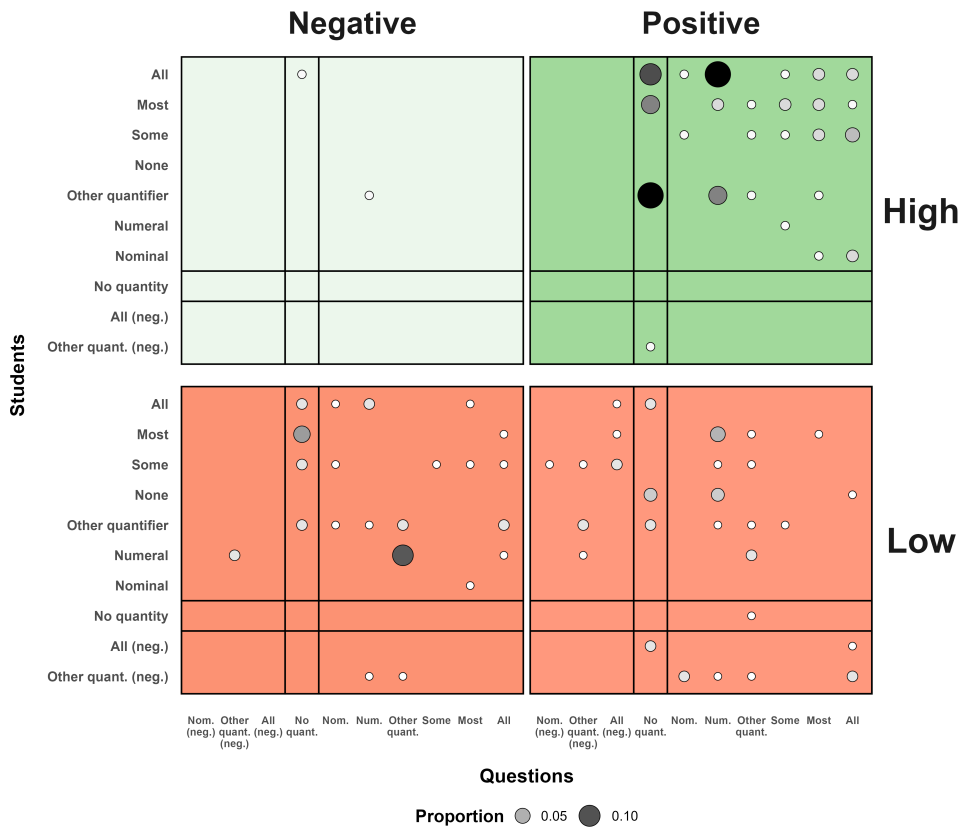

Figure 9.: Proportion of responses containing one or more quantifier, shown in terms of expression type and predicate polarity (**Positive** vs. **Negative**). The x-axis indicates expressions which refer to the questions whereas the y-axis indicates expressions which refer to the students. Darker and larger circles indicate a higher relative frequency.

Non-quantitative strategies

There were 24 descriptions containing no quantity expression whatsoever (9% of the single-reference data). While these responses were not included in the main analyses above, they nonetheless reveal additional strategies participants employed in trying to describe the stimuli according to the task. These descriptions involve some vague representation of the results such as a vague reference to the quality of the outcome and either an underspecified reference to the students, as in (25) and (26), or a reference to the school as a whole, as in (27). Another strategy is not to refer to the outcome at all, as in (28).

(25) *The students answered the questions correctly*

(26) *Riverside has a horrible success rate*

(27) *Green valley is upping the game*

(28) *Students unexpectedly gained high marks on a very easy exam paper*

Multi-reference strategies

There were 64 descriptions containing multiple references to students and/ or questions (19% of the original data). While these responses were not included in the main analyses above given their formal and pragmatic complexity, much like the non-quantitative strategies just discussed, they reveal additional strategies participants used in trying to describe the stimuli according to the task. The two most common forms of multiple reference include referring to particular students without resorting to any quantity expression (13 observations), as in (29) and (30) below, and using a sentence/ clause containing quantity expressions to modify or further specify another sentence/ clause containing quantity expressions (13 observations), as in (31) and (32) below.

(29) *According to the answer table Julian, Julia and Tim have done well*

(30) *According to the table Rebecca and Amy are good for nothing*

(31) *All of our students got over 3 quarters of the questions correct, some even got every question correct*

(32) *Almost half the students failed to answer a single question correctly, and another achieved a score well-below 50%*

3.2.2. Argumentative difficulty

Figure 10 shows the probability of each quantifier from Experiment 1, namely *all*, *most*, *some*, and *none*, being used to refer to students as a function of the difficulty metric introduced in section 2.2.2. The figure shows that, when referring to students in the **High** condition and framing their description positively, participants were more likely to use *all* at around 0.0 difficulty. From around 0.0 difficulty there was a steady increase in the probability of *most* being used, to the detriment of *all*. From stimuli with 0.5 difficulty or more, *some* became the most likely choice. Participants very rarely used any quantifier to refer to the students in the **High** condition while framing their description negatively, which is why that quadrant is left blank.

When referring to students in the **Low** condition and framing their description negatively, participants were more likely to use *all* at around 0.0 difficulty. From around 0.0 difficulty there was a steady increase in the probability of *most* being used, to the detriment of *all*. From around 0.35 difficulty participants only used *some*. When referring to students in the **Low** condition and framing their description positively, using *most* was the most likely choice at the [0.0, 0.5] interval. From stimuli with at least 0.5 difficulty, *all* became the most likely choice.

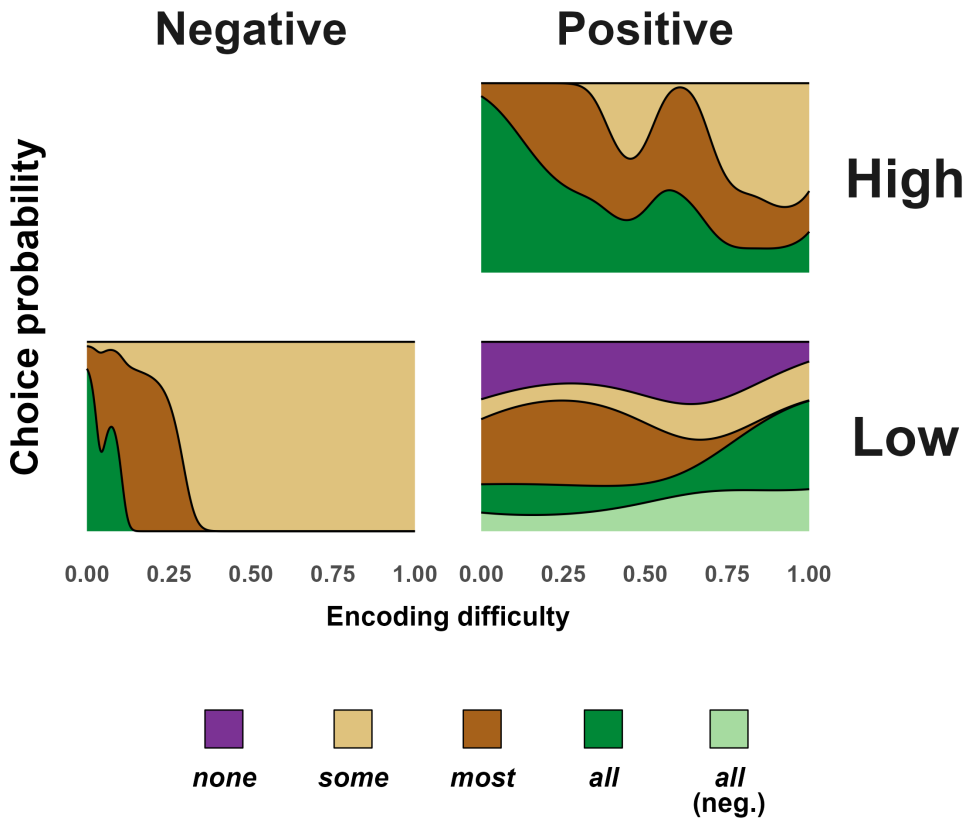

Figure 10.: Probability of *all*, *most*, *some*, or *none* being used to refer to students in Experiment 2 as a function of argumentative difficulty, shown in terms of the experimental conditions (**High** vs. **Low**) and predicate polarity (**Positive** vs. **Negative**). The x-axis indicates the varying degrees of difficulty whereas the y-axis indicates the probability of any given strategy being used.

Much like Fig. 10, Figure 11 shows the probability of each quantifier from Experiment 1 being used to refer to questions as a function of the difficulty metric. The figure shows that, when referring to students in the **High** condition and framing their description positively, participants were more likely to use *most* at around 0.0 difficulty.

At around 0.5 difficulty *all* was the most likely choice, from around 0.50 difficulty onwards *some* being the most likely choice, to the detriment of both *most* and *all*. Participants never used any quantifier to refer to the questions in the **High** condition while framing their description negatively, which is why that quadrant is left blank.

When referring to questions in the **Low** condition and framing their description negatively, *all* was the most likely choice around at the (0.0, 0.25) interval. From stimuli above 0.25 difficulty, *most* became the most likely choice. When referring to questions in the **Low** condition and framing their description positively, participants used *all* and *all* in a negated predicate equally likely.

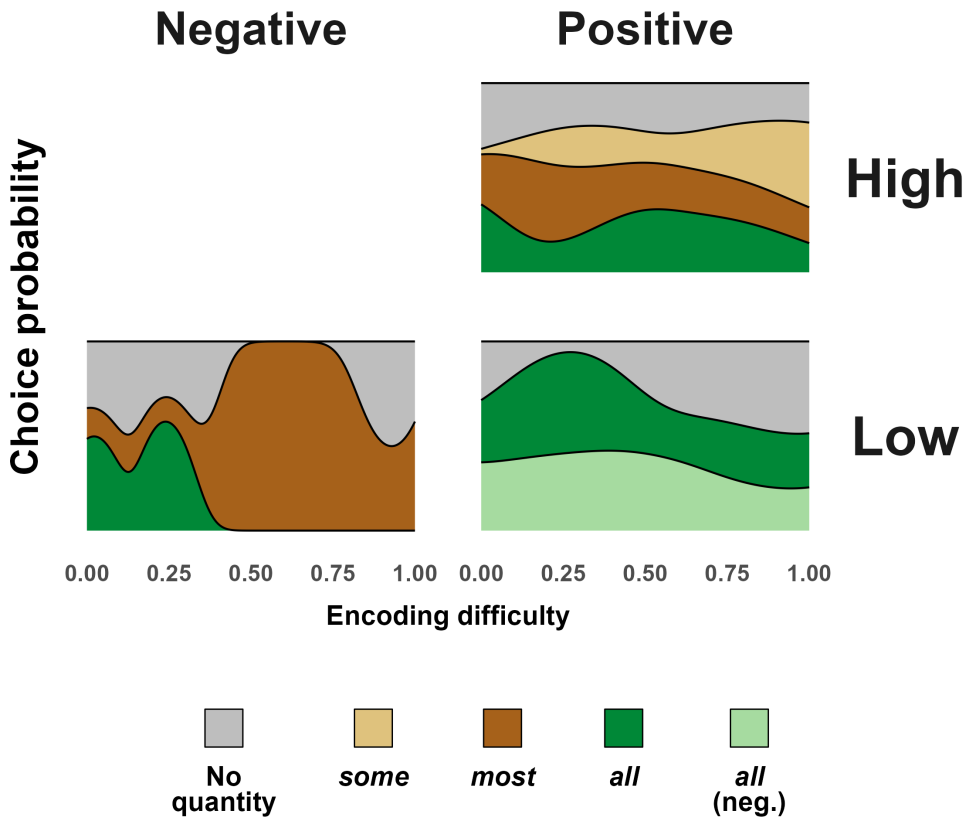

Figure 11.: Probability of *all*, *most*, *some*, or *none* being used to refer to questions in Experiment 2 as a function of argumentative difficulty, shown in terms of the experimental conditions (**High** vs. **Low**) and predicate polarity (**Positive** vs. **Negative**). The x-axis indicates the varying degrees of difficulty whereas the y-axis indicates the probability of any given strategy being used.

3.3. Discussion

Unlike in Experiment 1, where participants' task was to describe the stimuli by selecting between expressions made available to them as part of the task, in Experiment 2 they had to produce their own descriptions of the stimuli. As such, they could choose to frame their utterances relying on any strategy they deemed most appropriate, such as by using adjectives (e.g., "Majority of our students got answers *correct*"), as in Experiment 1, but also adverbs (e.g., "The students answered the questions *correctly*") and verbs (e.g., "some of the students *failed* completely"), or by avoiding overt polarity markers altogether and relying primarily on their usage of numerals (e.g., "None of the students scored above *75%*"). The results show that, in the **High** condition, people choose to frame their descriptions virtually always as a positive outcome, as in "All students got at least 75% *correct*". Interestingly, in the **Low** condition there was no bias for framing descriptions as a negative outcome, such that only about half of them are framed negatively, as in "over half of students got *0%*". This finding seems to provide partial evidence for the assumption that people prefer encoding favorable outcomes relative to unfavorable ones, in particular under the interpretation that they do so in ways which are favorable to them given their personal agenda. However, in the case of the more naturalistic utterances at hand, while people do prioritize outcomes which are favorable to them, at the same time they also seem to have a preference for referring to outcomes which are objectively positive, even if that is not in line with their argumentative goals, as evidenced by the fact that in **Low** condition about half of their descriptions are framed positively.

As for people's quantitative strategies in describing the stimuli, the results show that the most prevalent strategy involves using quantifiers, either only to refer to the students or to refer to both the students and the questions. That is followed by using a quantifier to refer to the students together with a numeral referring to the questions or using two numerals. 54% of all utterances containing quantifiers involve either *all* or *most*. But why do people tend to use these two quantifiers in particular? And when do they choose to combine them with other quantity expressions, including other quantifiers? As discussed in the introduction, it seems reasonable to assume that, given the possibility to do so, people use the strongest possible expression, such as *all*

relative to *most* and *some*.

In section 3.2.3 we addressed the issue of how the difficulty in describing particular stimuli in either the **High** or **Low** experimental condition affects people's choices of referential strategies. Much like in Experiment 1, we found that people tended to use informationally weaker quantifiers as the argumentative difficulty increases, at least when framing the exam outcomes in line with the respective experimental conditions. While utterances containing *all* are, in absolute terms, the most prevalent strategy involving quantifiers, people are more likely to shift towards using *most* and even *some* as argumentative difficulty increases. However, this pattern only characterized those conditions where the framing chosen by the participants (e.g., describing things in terms of "correct" responses) matches that of the condition (i.e., **High**). When this was not the case, results were rather unsystematic, with a fairly constant number of all responses regardless of argumentative difficulty.

4. General discussion

The results of both experiments show that people have clear preferences when using quantifiers to describe quantitative outcomes such as the results of school exams. In Experiment 1, where participants' choices were constrained to the quantifiers *all*, *most*, *some*, and *none*, they were shown to have a preference for using *all*, *most*, and *some* over *none*, regardless of how they chose to frame the exam outcomes. In Experiment 2, these biases were largely replicated with more naturalistic utterances, with *all* and *most* being preferred over *some*. Even though *some* was not as frequent a strategy as *all* and *most*, when used to refer to students in the **Low** condition it was combined with a wider range of expressions compared to other quantifiers. In that sense, likely due to its underinformative character, *some* lends itself to being combined with a wide range of expressions, including negated expressions, whereas *all* and *most* are used mostly with vague predicates which do not include any quantity expressions or with numerals.

Moving beyond absolute preferences, we found that, when taking the difficulty of describing a particular stimulus in a particular experimental condition into account,

there are rather systematic patterns in how the quantifiers are used: as the argumentative difficulty increases, so does the likelihood of *most* being used over *all*, as does the likelihood of *some* being used over *most* at higher degrees of difficulty. In other words, the likelihood of informationally weaker expressions being used increases with an increase in argumentative difficulty, suggesting that people opt for increasingly less informative expressions when it is harder to straightforwardly represent an outcome in terms of either a high or low success rate.

More generally, the data from Experiment 2 shows that using a quantifier to refer to students combined with either no quantity expression at all (e.g., "*Most* of the students got full marks") or a numeral referring to questions (e.g., "over half of students got *0%*) are particularly prevalent referential strategies. Here, one can only speculate about what constrains the choice of combining a quantifier with either one of the other two expression types: both vague predicates not involving quantity expressions and numerals can be used to frame an outcome as better or worse than it actually is, even though, at face value, numerals provide rather concrete information. For instance, in the utterance "Most of the students got full marks", which contains the adjective *full* modifying the noun *marks*, the framing seems to suggest that the outcome was quite good. While, at face value, this might be a straightforward description of an objectively good outcome, it might, perhaps, also be a strategic description of an excellent outcome where the producer is attempting to downplay the results, for instance when 4 out 5 students got full marks and a stronger description such as "Almost all students got full marks" might have been more informative.

In a similar vein, in the utterance "over half of students got 0%", the numeral *0%* seems to suggest that the outcome was quite bad. As with the choice of the adjective *full* above, this might be a straightforward description of an objectively bad outcome but it might also be a strategic description of an otherwise not particularly bad outcome, such as when 3 out of 5 students got a score of 0% but maybe 2 out of 5 got a score of 100%. While not per se a particularly good outcome, if one were to frame it positively instead of negatively, a different numeral would likely be used instead, e.g., "Almost half of students got 100%", or perhaps an altogether different quantifier combined with *0%*, e.g., "Not all students got 0%". In a nutshell, taking

the quantifiers as anchors to other expressions denoting quantity in an utterance, the choice of an accompanying expression type seems to be closely tied with the choice of a particular quantifier, such that choosing either an adjective or a numeral might induce qualitatively different interpretations depending on the polarity of the construction involving the quantifier, which may also involve modifiers such as *almost* and *over*. What's interesting is that even expressions which, at face value, might seem to signal rather concrete and, as such, informative values, such as a round numeral like *75%*, may in fact be playing a more vague role in communicating that a certain proportional threshold, as opposed to a particular exact value, has been reached, e.g., when a producer uses *75%* to emphasize that more than half of the questions have been answered correctly as opposed to simply signaling the exact proportion of correct questions.

From an encoding perspective, it remains an open question how language users decide on particular anchors when producing utterances containing quantity expressions, a question which is ultimately tied to what information they choose to communicate in the first place, as argued by Moxey and Sanford (2000) and Krifka (2005). Crucially, however, the current results highlight the fact that the choice of any given quantifier is modulated by the more general choice of how to frame one's utterance: positive framings in the **High** condition involve straightforward usages of *all* and *most*, whereas the patterns attested for the positive and negative framings in the **Low** condition are rather complementary to one another and involve a higher relative diversity of combinations of quantifiers with different accompanying expression types.

As discussed earlier, based on the findings by Cummins and Franke (2021), we put forward the assumption that language users generally have a preference to encode favorable outcomes over unfavorable ones. However, a competing set of biases which may counteract this preference are related to the well-attested positivity bias in language (i.e., Dodds et al., 2014; Warriner & Kuperman, 2015), and similarly the addition bias (Winter et al., 2023), as well as a similar bias to prefer risks being framed in terms of gains or survival as opposed to loss or mortality (i.e., Armstrong et al., 2002; Nabi et al., 2020). Our current findings seem to suggest that, in the production of utterances containing quantifiers, both of these competing preferences are at play:

on the one hand, producers do tend to prioritize communicating outcomes which are favorable to them given their argumentative goals; on the other hand, however, this bias is counteracted by a bias to frame situations in terms of objectively positive outcomes. Crucially, this has implications for the linguistic choices producers make when having different argumentative goals in mind: when their goal is to frame a situation as a positive outcome, there is a strong bias to produce utterances with a positive polarity; instead, when their goal is to frame a situation as a negative outcome, there is no clear bias to produce utterances with any particular polarity. These findings seem to support the argument put forward by Krifka (2005) that language users prefer communicating representations which are cognitively more salient, doing so using simpler linguistic forms whenever possible. Given the general picture suggesting that positive outcomes are more salient than negative ones, the findings are in line with the assumption above, such that, even when having an incentive to frame an outcome as negative, people still produce utterances with a positive polarity about half of the time. Interestingly, our results show that this seems to come at the price of having to use more complex constructions, such as constructions involving negative quantifiers (e.g., "not all the students answered all of the questions correctly"), unlike in the case where the utterances as a whole are framed negatively, where the choices tend to be more straightforward (e.g., "Some students made mistakes").

The current findings might also be said to support the assumption that people tend to use the best available reference point when communicating a quantitative outcome, an assumption which was also derived from the findings by Cummins and Franke (2021). In this regard, the systematic patterns found in the choice of quantifiers suggest precisely that, as a situation becomes increasingly less favorable to the producer, the usage of informationally weak quantifiers becomes increasingly more likely. All in all, these findings suggest that English speakers use quantifiers strategically when communicating quantitative information. Their choices are guided by argumentative considerations which take the difficulty of arguing for particular outcomes truthfully into account. Ultimately, though, the picture that emerges from comparing the constrained choices from Experiment 1 with those produced in Experiment 2 is that quantifiers are used in conjunction with a wider range of linguistic strategies avail-

able to speakers of English, including other lexical expressions of quantity which may serve as salient competitors to any given quantifier. One interesting finding is that *some* seems to lend itself particularly well to being combined with other quantifiers, whereas *all*, *most* and even *none* are more often produced in combination with either no quantity expression at all or with numerals.

Future research should aim to investigate more explicitly not only what motivates producers to make any non-trivial choices between different types of quantity expressions but also what biases affect people's encoding of a situation in terms of different affective polarities and how that interacts, more generally, with the process of utterance formulation. In this sense, data and analyses such as the ones reported in the present study provide pointers to what expressions might be more salient during production and what comparison sets are more likely to be activated in interpretation.

Indeed, one particularly exciting question for future research relates exactly to the interpretation of utterances like the ones attested in our data set: how do interpreters make sense of otherwise informationally similar utterances containing different types of quantity expressions? In other words, moving beyond the question of how interpreters make sense of the quantities encoded in utterances containing quantity expressions, which has been traditionally at the center of studies focused on quantifier interpretation, one might ask, instead, how the combination of any particular quantifier with other more or less vague linguistic forms impacts interpreters' qualitative understanding of a particular state of affairs as good or bad. For instance, one might ask to what extent a producer might be perceived as lying given their choice to combine a quantifier with either a numeral or a vague predicate not containing any quantity expression. Given that producers' choices generally reflect biases related to how particular audiences might interpret their utterances, one might ask if it is reasonable to assume that language users who are aware of potentially misleading argumentative goals interpret 'objective' and concrete expressions like numerals in straightforward informative ways. Might their interpretation instead be constrained by biases related not only to the informativity of particular quantity expressions but maybe also to the trustworthiness of an utterance as a whole given a producer's choice of particular linguistic forms?

Despite shining light onto the question of how language users modulate their linguistic behavior in light of argumentative considerations, one limitation of our current setup is that various types of complexity are entwined into what we call argumentative difficulty. Indeed, the difficulty associated with describing a particular stimulus in our task as either a good or bad outcome is dependent not only on how informative one can be but also on how the stimulus looks like, i.e., the exact distribution of green to red cells in the images, as well as on how easy it might be to encode something in the first place, e.g., a stimulus which is highly difficult to describe might, by default, demand more verbose descriptions, which in turn are likely to correlate with linguistically more complex utterances. Future work could look to disentangle these different dimensions from one another, for example by explicitly manipulating the complexity of the visual stimuli. Ultimately, our current results illustrate that combining highly controlled experimentation with less constrained production tasks can provide rich insights into the pragmatic biases underlying language usage, which, in turn, can help constrain abstract theorizing by grounding it in more naturalistic and plausible usage patterns.

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Disclosure statement

An unnumbered section, e.g. `\section*{Disclosure statement}`, may be used to declare any potential conflict of interest and included *in the non-anonymous version* before any Notes or References, after any Acknowledgements and before any Funding information.

Data availability statement

All materials, data, and analysis scripts can be found in the following repository: `https://tinyurl.com/yrasxa9r`.

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